Nitin Kumar Bairwa

nitinkumarbairwa101@gmail.com | +91 9588867395 | Nitin Kumar
nitin-kumar101

Academic Qualifications

Year	Degree/Certificate	Institute	CGPA/%
2024	M.SC(5YR)	Indian Institute of Technology, Kharagpur	8.02/10
2019	RBSE(XII)	Sumit Bal Sr. Sec. School, Kota	91.80%
2017	RBSE(X)	Sumit Bal Sr. Sec. School, Kota	93.83%

Experience

TATA CONSULTANCY SERVICES | AI Engineer (***Aug'24 - Current***)

Enterprise GenAI Platform Development

- Collaboratively contributed to a globally distributed team in developing an AI platform for a major client's employees
- Built AI plugins with RAG, agentic workflows, multimodal, multilingual processing using **Azure OpenAI** and **AWS Bedrock**
- Designed document pipelines with **LangChain** for ingestion, retrieval, and transformation, and **LangGraph** for agentic workflows
- Deployed **FastAPI** pipelines with PostgreSQL (**pgvector**), vector and document DBs for metadata indexing and semantic search
- Developed a plugin registry in FastAPI to manage plugin types, parameters, and prompts with dynamic API endpoint execution
- Implemented document processing pipelines for multilingual text, images, diagrams, and large PDF files **exceeding 3,000 pages**
- Engineered ingestion of diverse formats (PDF, PPTX, Excel, MHTML, PNG, JPG) into **PostgreSQL** with the pgvector extension
- Built processing and retrieval pipelines for complex circuit diagrams related files, enabling extraction of component relationships
- Developed **MCP** tools for custom retrieval and validation, integrated into the core RAG agent for improved performance

Virtual Sales Agent

- Developed Generative AI–based multi-agent implementation for a major global printer manufacturing enterprise client project
- Implemented a multi-agent RAG application for sellers to enhance customer engagement and enable upsell with **Azure AI**
- Built specialized **Azure AI Search** retriever agents and API-calling agents for accurate information retrieval and automation
- Leveraged Microsoft's **MCP** framework to host reusable prompts and tools that optimized performance of intelligent AI agents
- Designed condition-based cyclic workflows with **LangGraph** for orchestrating agent responses and improving system adaptability
- Implemented persistent memory for conversational agents in **Cosmos DB** to manage multiple user sessions and interactions

Projects

Extracting Material Parameters using ANNs | Master's Thesis Project (***April'24***)

- Developed ANNs for THz-TDS parameter extraction, achieving mean squared error significantly lower than traditional methods
- Implemented **DNNs** and **CNNs** effectively employing **scaled conjugate gradient backpropagation** and **Adam** optimizers
- Achieved remarkable MSE scores of **0.000012** and **0.000015** for DNN and CNN, respectively, on simulated data with noise

Keyword Extraction from Road Accident Tweets | Natural Language Processing (***May'23 - July'23***)
Advisor: Professor Sobhan Sarkar,IIM Ranchi

- Analyzed road accident tweets with advanced NLP techniques, extracting **200,000+** tweets utilizing **Snscrape**
- Utilized few-shot classification enhanced by **INSTRUCTOR** embeddings with **K-means** clustering using cosine distance metrics
- Innovated keyword extraction, blending topic models (LDA,BERTopic) with word embeddings like **RoBerta**
- The proposed model demonstrated strong coherence scores **($>$ 0.8)**, signifying its effectiveness across various topic numbers

2D Runner Game using Pygame | Python Development ***GitHub Link***

- Developed a Python game using pygame, showcasing character animations, evolving backgrounds, and dynamic gameplay
- Strategically integrated coin accumulation, obstacle avoidance, and immersive visuals for a truly engaging player experience

Competitions/Conferences

Inter- Hall Data Analytics Challenge — IIT Kharagpur (***Feb'22 -March'22***)
Objective: *Analyze 38 months of sales data for an electronics hardware company and predict upcoming month's sales*

- Conducted thorough Exploratory Data Analysis on the time series data, using statistical tests such as **T-test and Z-test**
- Identified anomalies using **Inter Quantile Range, Isolation Forest**, and subsequently replacing them with moving average values
- Employed time series forecasting models **(Exponential Smoothing, ARIMA, SARIMA, Holts Winter)** and DL models **(LSTMs)**
- Achieved **0.18 MAPE** by constructing a customized ensemble model of 8 forecasting algorithms, ensuring robust sales prediction

Skills and Expertise

- **Programming Languages:** C | C++ | **Python** | HTML | CSS | JavaScript | **SQL**
- **Softwares** & **Frameworks:** Matplotlib | **Pandas** | **NumPy** | NLTK | TensorFlow | LangChain | **scikit-learn** | Scipy | BeautifulSoup | **Keras** | **Langgraph** | **FastAPI** | **Flask** | Kubernetes

Certifications

- **Microsoft Certified:** Azure Fundamentals, Azure AI Engineer Associate, Azure AI Fundamentals
- **CourseWork:** Generative AI with Large Language Models, Neural Networks and Deep Learning, Probability and Statistics, Machine Learning foundations and applications, Programming and Data Structures